A PRODUCT-CUBE COUNTEREXAMPLE TO THE DENSITY ASSD2P QUESTION

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Status

This packet records a likely-valid counterexample to the second question in Question 3.2 of Ciaci's paper *On the transfinite symmetric strong diameter two property* [1]. It gives a compact Hausdorff, hence T_4 locally compact, space K with $\mathrm{d}(K) > \aleph_0$ such that $C_0(K) = C(K)$ fails 1-ASSD2P$_{\mathrm{d}(K)}$. Thus the proposed density-only positive statement for 1-ASSD2P is false.

The packet does not answer the first question, namely whether $C_0(X)$ must fail SSD2P$_{\mathrm{c}(X)^+}$ for every T_4 locally compact space X.

Source question

The source asks whether Theorem 3.1 can be rewritten using only one cardinal function:

> Let X be a T_4 locally compact space. Is it true that $C_0(X)$ fails the SSD2P$_{\mathrm{c}(X)^+}$?
> Is it true that $C_0(X)$ enjoys the 1-ASSD2P$_{\mathrm{d}(X)}$, whenever $\mathrm{d}(X) > \aleph_0$?

> It remains unclear whether the statement of Theorem 3.1 can be written using only one cardinal function. Namely, we don't know the answer to the following two questions.
>
> **Question 3.2.** *Let X be a T_4 locally compact space. Is it true that $C_0(X)$ fails the* SSD2P$_{\mathrm{c}(X)^+}$*? Is it true that $C_0(X)$ enjoys the* 1-ASSD2P$_{\mathrm{d}(X)}$, *whenever* $\mathrm{d}(X) > \aleph_0$*?*

Definitions used

All Banach spaces are real, as in the source paper. For an infinite cardinal κ, a Banach space E has 1-ASSD2P$_\kappa$ if, for every set $A \subset S_{E^*}$ with $|A| < \kappa$, there are a set $B \subset S_E$ norming A and a vector $y \in S_E$ such that $B \pm y \subset S_E$. Here B norms A means that for every $a^* \in A$ and every $0 < r < 1$ there is $b \in B$ with $a^*(b) \geq r$.

Idea of the proof

A strictly positive probability measure blocks the exact symmetric 1-ASSD2P requirement. If $B \pm y$ stays in the unit sphere of $C(K)$, then for every $f \in B$ the pointwise inequality

$$|f(t)| + |y(t)| \leq 1$$

holds. A full-support measure μ sees every nonzero continuous y, so $\int |y|\, d\mu > 0$. Consequently every such f satisfies

$$\mu(f) \leq 1 - \int |y|\, d\mu < 1,$$

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and B cannot norm the singleton $\{\mu\}$.

The counterexample

Theorem 1. *Let Γ be a set with $|\Gamma| > \mathfrak{c} = 2^{\aleph_0}$, and put $K = [0,1]^\Gamma$ with the product topology. Then K is a T_4 locally compact space with $\mathrm{d}(K) > \aleph_0$, but $C_0(K) = C(K)$ fails 1-ASSD2P$_{\mathrm{d}(K)}$.*

Proof. The space K is compact Hausdorff by Tychonoff's theorem. In particular it is normal and locally compact, and $C_0(K) = C(K)$.

First, K is nonseparable. Let $D \subset K$ be countable. For each $\gamma \in \Gamma$, define

$$v_\gamma = (x(\gamma))_{x\in D} \in [0,1]^D.$$

Since D is countable, $|[0,1]^D| = \mathfrak{c}$. Since $|\Gamma| > \mathfrak{c}$, there are distinct $\gamma, \eta \in \Gamma$ such that $v_\gamma = v_\eta$. Hence $x(\gamma) = x(\eta)$ for every $x \in D$. Thus D is contained in the closed proper subset

$$F_{\gamma,\eta} = \{z \in K : z(\gamma) = z(\eta)\}.$$

The set is proper because points with different γ and η coordinates exist. Therefore D is not dense in K. No countable subset of K is dense, so $\mathrm{d}(K) > \aleph_0$.

Let μ be the product of Lebesgue probability measures on the factors $[0,1]$. This regular Borel probability measure has full support: every nonempty basic open cylinder restricts only finitely many coordinates and has positive product measure.

Suppose, for contradiction, that $C(K)$ has 1-ASSD2P$_{\mathrm{d}(K)}$. Since $\mathrm{d}(K) > \aleph_0$, the singleton $A = \{\mu\} \subset S_{C(K)^*}$ has cardinality $< \mathrm{d}(K)$. Hence there are $B \subset S_{C(K)}$ norming A and $y \in S_{C(K)}$ such that $B \pm y \subset S_{C(K)}$.

Fix $f \in B$ and $t \in K$. Because both $f + y$ and $f - y$ are in the unit sphere of $C(K)$, their sup norms are at most one. Therefore

$$|f(t) + y(t)| \le 1, \qquad |f(t) - y(t)| \le 1.$$

For real scalars a, b,

$$\max\{|a + b|, |a - b|\} = |a| + |b|.$$

Applying this with $a = f(t)$ and $b = y(t)$ yields

$$|f(t)| + |y(t)| \le 1.$$

In particular $f(t) \le |f(t)| \le 1 - |y(t)|$ for every $t \in K$. Hence

$$\mu(f) = \int_K f\,d\mu \le \int_K (1 - |y|)\,d\mu = 1 - \int_K |y|\,d\mu.$$

Since $y \in S_{C(K)}$, it is not identically zero. Choose t_0 with $|y(t_0)| > 0$. By continuity, $|y| \ge \delta > 0$ on some nonempty open neighborhood U of t_0. Full support gives $\mu(U) > 0$, so $\int_K |y|\,d\mu \ge \delta\mu(U) > 0$.

Thus there is a constant $q < 1$ such that $\mu(f) \le q$ for all $f \in B$. This contradicts the assertion that B norms $\{\mu\}$. Therefore $C(K)$ fails 1-ASSD2P$_{\mathrm{d}(K)}$. □

Novelty and verification notes

- Local run indexes were searched for `2302.14414`, the paper title, and the core phrases `ASSD2P density cellularity`; no exact duplicate solution packet or attempt answering this question was found.
- Bounded web search on 2026-06-29 used the arXiv id, the title, and queries such as "$C_0(X)$ ASSD2P density" and "transfinite symmetric strong diameter two property density cellularity". The search surfaced the source arXiv page but no later exact answer or this counterexample.

- The proof uses only standard facts: Tychonoff compactness, the usual product probability measure on a compact product of intervals, and the Riesz–Markov identification of $C(K)^*$ with regular Borel measures.

Human review recommendation

Review as a likely-valid counterexample to the second half of Question 3.2. The key points to check are:

- the source definition of 1-ASSD2P$_\kappa$ requires testing all $A \subset S_{C(K)^*}$ of cardinality $< \kappa$, so the singleton $\{\mu\}$ is admissible when $\mathrm{d}(K) > \aleph_0$;
- the product Lebesgue measure on $[0,1]^\Gamma$ is regular and strictly positive on nonempty open subsets;
- the scalar identity $\max\{|a+b|, |a-b|\} = |a| + |b|$ is the only place where the real-scalar convention is used.

References

[1] S. Ciaci, *On the transfinite symmetric strong diameter two property*, arXiv:2302.14414, 2023. `https://arxiv.org/abs/2302.14414`